ASSIGNMENT ON K-MEANS CLUSTERING.
Problem Statement:
Implement K-Means clustering/ hierarchical clustering on sales_data_sample.csv dataset.
Determine the number of clusters using the elbow method.
Dataset link : https://www.kaggle.com/datasets/kyanyoga/sample-sales-data

import os #These two lines are used to limit the number of CPU cores that a Python library can use. It tells Loky to use a maximum of 4 CPU cores when determining its available CPU count.

 It can help avoid CPU-count detection warnings in some environments.

```
os.environ["LOKY_MAX_CPU_COUNT"] = "4"

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

from sklearn.preprocessing import StandardScaler

from sklearn.cluster import KMeans

df = pd.read_csv("sales_data_sample.csv", encoding="ISO-8859-1")

print("Dataset loaded successfully!")

print(df.head())

print("Shape of dataset:", df.shape)


print("\nColumn names:")

print(df.columns)


print("\nMissing values:")

print(df.isnull().sum())

# Fill missing postal codes

df['POSTALCODE'] = df['POSTALCODE'].fillna('Unknown')


# Fill missing categorical values with mode

df['STATE'] = df['STATE'].fillna(df['STATE'].mode()[0])


df['TERRITORY'] = df['TERRITORY'].fillna(df['TERRITORY'].mode()[0])
```

```
# Fill missing address with Unknown
df['ADDRESSLINE2'] = df['ADDRESSLINE2'].fillna('Unknown')

# Check missing values
print(df.isnull().sum())
X = df[[
   "QUANTITYORDERED",
   "PRICEEACH",
   "SALES",
   "MSRP"
]]

print("Selected features:")
print(X.head())

print("\nMissing values in selected features:")
print(X.isnull().sum())
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)

print("Feature scaling completed!")
print(X_scaled[:5])
from sklearn.cluster import KMeans

wcss = []

for i in range(1, 11):
   kmeans = KMeans(n_clusters=i, random_state=42, n_init=10)
```

```
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

print("WCSS values:")
print(wcss)
import matplotlib.pyplot as plt

plt.figure(figsize=(8, 5))

plt.plot(range(1, 11), wcss, marker='o')

plt.xlabel("Number of Clusters (K)")
plt.ylabel("WCSS")
plt.title("Elbow Method")

plt.grid()
plt.show()
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)

clusters = kmeans.fit_predict(X_scaled)

print("K-Means clustering completed!")
print("First 20 cluster labels:")
print(clusters[:20])
df["Cluster"] = clusters

print(df[[
    "QUANTITYORDERED",
    "PRICEEACH",
    "SALES",
    "MSRP",
```

```
    "Cluster"
]].head(10))
print("Number of records in each cluster:")
print(df["Cluster"].value_counts().sort_index())
centers = scaler.inverse_transform(kmeans.cluster_centers_)
centers_df = pd.DataFrame(
    centers,
    columns=["QUANTITYORDERED", "PRICEEACH", "SALES", "MSRP"]
)
print("Cluster Centers:")
print(centers_df)
plt.figure(figsize=(8, 5))
plt.scatter(
    df["QUANTITYORDERED"],
    df["SALES"],
    c=df["Cluster"]
)
plt.xlabel("Quantity Ordered")
plt.ylabel("Sales")
plt.title("K-Means Clustering of Sales Data")
plt.grid()
plt.show()
df.to_csv("sales_data_clustered.csv", index=False)
print("Clustered dataset saved successfully!")
```

Unsupervised Machine Learning is a type of machine learning in which the model learns patterns and relationships from data without using predefined labels or output values.

In simple words, we give the machine a dataset, but we don't tell it which data belongs to which category. The algorithm automatically discovers patterns or groups.

1. Clustering

Groups similar data points together.

Examples:

- K-Means Clustering,Hierarchical Clustering,DBSCAN

- • K-Means requires us to specify the number of clusters (K).
- • Hierarchical Clustering creates a tree-like structure called a dendrogram, which helps decide the number of clusters.
- DBSCAN is a density-based clustering algorithm that groups closely located data points and identifies outliers or noise.
- Q5. What is PCA? 2.dimensionality reduction
- PCA stands for Principal Component Analysis. It reduces the number of features in a dataset while retaining as much important information as possible.

- 3. 2. Association

Finds relationships between different items in a dataset.

Examples:

- Apriori Algorithm
- FP-Growth Algorithm

Real-life example: If customers buy bread, they may also buy butter. This helps supermarkets understand purchasing patterns.

K-Means Clustering is an unsupervised machine learning algorithm used to divide data into K different groups (clusters) based on similarities.

- K represents the number of clusters.
- Each cluster has a centroid (center point).
- Data points are assigned to the nearest centroid.
- The algorithm repeatedly updates centroids until the clusters become stable.

Elbow Method:

The Elbow Method is used to find the optimal number of clusters.

It calculates WCSS (Within-Cluster Sum of Squares) for different values of K. The optimal K is selected at the point where the WCSS reduction starts slowing down, forming an elbow-like shape.

encoding tells Python how to read and interpret the characters stored in a CSV file latin1 = ISO-8859-1Both support Western European characters, including accented letters.Both help read CSV files containing special characters without encoding errors.

Why does WCSS decrease when K increases?

Answer: When the number of clusters increases, data points are divided into smaller groups. This reduces the distance between data points and their respective cluster centroids, resulting in lower WCSS.

The K-Means algorithm divided the sales dataset into three clusters. A total of 2,823 records were grouped based on Quantity Ordered, Price Each, Sales and MSRP.

The cluster distribution is:

- Cluster 0: 975 records
- Cluster 1: 852 records
- Cluster 2: 996 records

The cluster centers represent the average values of the selected features in each cluster.

Result:

K-Means clustering was successfully implemented on the sales dataset. The Elbow Method was used to select three clusters, and the sales records were grouped into three clusters based on their similarities.

5. Important viva questions

What is a cluster center? They help us understand the characteristics and average values of each cluster.

A: It is the centroid or average position of all data points belonging to a particular cluster.

 Cluster 0: Lower-priced products with relatively low sales.

 Cluster 1: Higher quantities ordered and the highest average sales.

 Cluster 2: Lower quantities ordered than the other groups, with relatively high prices and moderate sales.

The WCSS value decreases as the number of clusters increases. From the Elbow Method graph, K = 3 appears to be a reasonable choice for the number of clusters.

Conclusion: The optimal number of clusters is selected as 3, and K-Means clustering is performed using three clusters.

Postal codes are identifiers rather than numerical measurements. Therefore, calculating their mean is not meaningful. I filled missing postal codes with Unknown.

K-Means is used in customer segmentation, product recommendation, banking, healthcare, traffic management and image segmentation.

K-Means is widely used in customer segmentation. For example, an e-commerce company can group customers based on their purchasing behaviour and spending patterns. This helps the company understand customer interests and create targeted marketing strategies."

Can K-Means detect fraud?

Ans: It can identify unusual transaction patterns by grouping similar transactions, but it cannot independently confirm fraud.

Q4. Why do companies use K-Means?

Ans: It helps companies discover hidden patterns in large datasets and group similar records for better decision-making.

The K-Means clustering algorithm was successfully implemented on the sales dataset. The Elbow Method was used to determine the suitable number of clusters. From the WCSS values and Elbow graph, the appropriate number of clusters was found to be **3**.

The dataset was divided into three clusters containing **975, 852, and 996 records** respectively. The clusters were formed based on Quantity Ordered, Price Each, Sales, and MSRP. Cluster 1 represents orders with relatively higher quantity and the highest average sales, while Cluster 0 represents relatively lower-price and lower-sales orders. Cluster 2 contains orders with lower average quantity but relatively higher prices.

Thus, K-Means successfully grouped the sales records into meaningful clusters based on their sales characteristics.

- **Clusters (\(K=3\)):** The data is segmented into three distinct groups represented by different colors:
    1. **Purple Cluster (Low Spend):** Represents orders with low sales value (generally under $4,000) regardless of the quantity ordered (stretching from low quantities up to around 65 units).
    2. **Yellow Cluster (Low-to-Medium Quantity, Medium Spend):** Consists of orders with low-to-moderate quantities (mostly between 15 and 40 units) but with a higher price point per item, resulting in sales values between roughly $2,000 and $5,500.
    3. **Teal Cluster (High Value / High Volume):** Contains the highest-value transactions, covering sales values from $4,000 up to $14,000+. This includes dense moderate-quantity orders as well as extreme high-quantity outliers (up to 100 units).

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encoding tells Python how to read and interpret the characters stored in a CSV file. **Why do we use encoding here?**

Sometimes CSV files contain special characters such as:

- é
- ñ
- ü

If Python reads the file using an incompatible encoding, it may produce a UnicodeDecodeError.